Data Requirements Checklist — BDBA Capstone

Module 1: Demographics

Panel 1.1 — U.S. Population by Single-Year Age, Projected 2025–2050

Variable	Detail
What	Population count by single-year age (0–100+), both sexes, for each projection year 2025–2050
Granularity	Single-year age × single calendar year
Format	CSV or Excel
Primary source	**Census Bureau National Population Projections** (2023 revision). Table: "Projected Population by Single Year of Age, Sex, Race, and Hispanic Origin: Main Series, 2024 to 2100"
URL	https://www.census.gov/data/tables/2023/demo/popproj/2023-summary-tables.html → download NP2023-T3 or the detailed files
Backup source	**UN World Population Prospects 2024** — provides single-year age projections for the U.S. under multiple fertility/mortality variants
Backup URL	https://population.un.org/wpp/Download/Standard/CSV/
Notes	Census provides a **main series** (single projection) and may also provide high/low immigration variants. For Monte Carlo, we may want to use Census main series as the central path and apply small perturbations rather than treating the Census projection itself as uncertain. The demographic structure (relative cohort sizes) is very stable — the people who will be 73 in 2038 are already alive today.
Monte Carlo treatment	Low uncertainty. Mortality is the main variable — use ±1–2 year life expectancy shifts across draws. Immigration adds minor uncertainty for younger cohorts.

Panel 1.2 — Historical U.S. Population by Age (for validation/backtesting)

Variable	Detail
What	Historical population by single-year age, 2000–2024
Primary source	**Census Bureau Population Estimates** (intercensal and postcensal)
URL	https://www.census.gov/programs-surveys/popest/data/tables.html

Variable	Detail
Notes	Useful for validating Module 2/3 against historical ICI fund flow data. Not strictly required for the forward simulation but makes backtesting possible.

Module 2: Inflows

Panel 2.1 — 401(k) Participation Rate by Age

Variable	Detail
What	Fraction of workers in each age bracket who participate in a 401(k) / DC plan
Granularity	Age brackets (ideally 5-year: 20–24, 25–29, ..., 60–64; or at least <25, 25–34, 35–44, 45–54, 55–64)
Format	PDF tables (will need manual extraction or OCR)
Primary source	**Vanguard "How America Saves" 2024** — published annually, based on ~5 million participants across ~1,900 plans
URL	https://institutional.vanguard.com/content/dam/inst/iig-transformation/has/2024/pdf/has-report.pdf (check for 2025 edition)
Backup source	**BLS National Compensation Survey** (Table 2: Retirement benefits), **EBRI Issue Briefs**
Backup URL	https://www.bls.gov/ncs/ebs/benefits/2023/employee-benefits-in-the-united-states-march-2023.pdf
Key numbers to extract	Overall participation rate (~80–85% with auto-enrollment), participation by age group, participation by income bracket
Notes	Madrian & Shea (2001) found auto-enrollment → ~85% participation. Vanguard's current data shows similar or higher. We use Vanguard as the primary calibration because their sample is large, recent, and directly from plan records (not surveys).
Monte Carlo treatment	Moderate uncertainty. Base rate ~83% ± 5% across draws. Age pattern (lower participation among young, higher among middle-aged) is stable.

Panel 2.2 — Contribution Rate (Deferral Rate) by Age

Variable	Detail
What	Average employee contribution as % of salary, by age group
Granularity	Same age brackets as participation rate

Variable	Detail
Primary source	**Vanguard "How America Saves" 2024 —** reports average deferral rates by age and income
Key numbers to extract	Average total savings rate (employee + employer) by age: typically ~6–7% for young workers, ~10–12% for 50+ (with catch-up contributions)
Notes	Also need the **employer match —** typically 3–5% of salary. Vanguard reports total savings rate (employee + employer). For the model, what matters is total dollars flowing into the plan, so total rate × salary = contribution.
Monte Carlo treatment	Moderate uncertainty. Deferral rates vary by income and plan design. Use distribution from Vanguard data.

Panel 2.3 — Median/Average Salary by Age

Variable	Detail
What	Earnings by age group to convert contribution rates to dollar amounts
Granularity	5-year age brackets (or single-year if available)
Primary source	**BLS Current Population Survey (CPS) —** Table: "Median usual weekly earnings of full-time wage and salary workers by age"
URL	https://www.bls.gov/cps/cpsaat39.htm
Backup source	**Social Security Administration Average Wage Index (AWI) —** gives national average wage by year, not by age
Backup URL	https://www.ssa.gov/oact/cola/AWI.html
Supplementary	**Census Bureau ACS —** provides income by age in finer brackets
Notes	We need age-specific earnings, not just national average. BLS CPS has this in 10-year age brackets. For finer granularity, the Census ACS PUMS microdata would allow custom tabulations but is complex. For the simulation, 5-year brackets interpolated to single years is sufficient.
Monte Carlo treatment	Low-moderate uncertainty. Nominal wage growth ~3–4%/year. Real earnings by age profile is stable.

Panel 2.4 — TDF Glide Path (Equity Allocation by Age)

Variable	Detail
What	Target equity allocation as a function of years to retirement (or participant age) for the major TDF families

Variable	Detail
Granularity	Annual or 5-year steps from age 25 to age 85+
Primary source	**Morningstar Target-Date Fund Landscape Report 2024 —** comprehensive survey of all major TDF families, with glide path charts
URL	Search Morningstar's research portal; also available via Morningstar Direct if you have institutional access through IE
Backup source	**Individual provider websites —** Vanguard Target Retirement Funds, Fidelity Freedom Funds, BlackRock LifePath — each publishes their glide path publicly
Backup URLs	Vanguard: https://investor.vanguard.com/investment-products/mutual-funds/target-retirement-funds → each fund's page shows the glide path. Fidelity: similar. T. Rowe Price: similar.
Key numbers	Typical glide path: ~90% equity at age 25, ~70% at 45, ~50% at 65, ~30% at 75+. "Through" funds continue de-risking post-retirement; "to" funds level off at retirement. Most large providers use "through."
Notes	Parker et al. (2023) provide aggregate glide path data in their paper (Figure 1 and Table I). You may be able to use their published numbers directly rather than sourcing from Morningstar.
Monte Carlo treatment	Low uncertainty for the simulation. The glide path is a design choice, not a random variable. Use the Vanguard glide path as the central case. Sensitivity: test with ±5% equity allocation shift.

Panel 2.5 — Number of Active 401(k) Participants / DC Plan Coverage

Variable	Detail
What	Total number of active participants in DC plans (401(k), 403(b), etc.) by year; ideally by age
Primary source	**ICI Factbook —** annual publication with total DC plan assets, contributions, and participant counts
URL	https://www.ici.org/system/files/2024-05/2024-factbook.pdf (check for 2025 edition)
Backup source	**DOL Form 5500 Summary —** aggregate data from all plan filings
Backup URL	https://www.dol.gov/agencies/ebsa/researchers/statistics/retirement-bulletins/annual-statistical-supplement
Notes	ICI reports ~70 million active 401(k) participants as of 2023. Total DC participants (including 403(b), 457, etc.) is higher. We need this to scale our per-person inflows to aggregate dollars.

Panel 2.6 — Total DC Plan Assets and Historical Fund Flows

Variable	Detail
What	Time series of total DC plan assets, annual contributions (inflows), and annual distributions (outflows), 2000–2024
Primary source	**ICI Factbook —** Chapter 8 (Retirement) provides annual aggregate data
Also	**Federal Reserve Z.1 Financial Accounts** (Table L.118.c for DC plans or Table L.227 for mutual funds)
URL	https://www.federalreserve.gov/releases/z1/
Notes	This is the **calibration anchor**. Our model should reproduce approximately correct aggregate inflows and outflows when fed historical demographics and parameters. If the model says 2020 net inflows were $X billion and ICI data says $Y billion, the discrepancy tells us whether our parameters are calibrated correctly.
Critical for	Backtesting / validation of Modules 2 and 3 before running forward projections.

Module 3: Outflows

Panel 3.1 — IRS RMD Distribution Period Table

Variable	Detail
What	The Uniform Lifetime Table from IRS Publication 590-B, giving the distribution period (divisor) for each age from 72 to 100+
Granularity	Single-year age
Primary source	**IRS Publication 590-B** (Distributions from Individual Retirement Arrangements)
URL	https://www.irs.gov/publications/p590b
Key numbers	Age 73: distribution period = 26.5 (withdraw ~3.8%). Age 75: 24.6 (~4.1%). Age 80: 20.2 (~5.0%). Age 85: 16.0 (~6.3%). Age 90: 12.2 (~8.2%).
Notes	**SECURE 2.0 age thresholds:** Born 1951–1959 → RMDs begin at age 73. Born 1960+ → RMDs begin at age 75. These are deterministic — no uncertainty needed. The distribution period table was updated in 2022 (longer periods = lower required withdrawals than the old table). Make sure to use the current (post-2022) table.

Variable	Detail
Monte Carlo treatment	None — this is deterministic. The *account balance* to which the RMD percentage applies is stochastic (depends on past returns).

Panel 3.2 — Average Retirement Account Balance by Age

Variable	Detail
What	Mean and median 401(k)/IRA balance by age bracket for current retirees and near-retirees
Granularity	5-year age brackets (55–59, 60–64, 65–69, 70–74, 75–79, 80+)
Primary source	**Vanguard "How America Saves" 2024** — reports average balance by age group
Backup source	**Federal Reserve Survey of Consumer Finances (SCF)** — triennial survey (2022 latest), detailed wealth data by age
Backup URL	https://www.federalreserve.gov/econres/scfindex.htm
Also useful	**Poterba, Venti & Wise** tables from the project papers — gives wealth decomposition for ages 65–69 (2008 HRS data, corrected 2013), but outdated in levels
Notes	Account balances have grown enormously since the Poterba/Venti/Wise 2008 snapshot due to the 2009–2024 bull market. Use Vanguard or SCF for current levels. Vanguard's data covers ~5 million participants, making it the best available source. The SCF is a nationally representative survey but is only triennial.
Monte Carlo treatment	Moderate uncertainty. Balances depend on past returns and contributions. Use SCF distribution to calibrate the mean and spread.

Panel 3.3 — Voluntary Withdrawal Rates by Age (Pre-RMD)

Variable	Detail
What	Fraction of account holders who take voluntary distributions before RMD age, and the average withdrawal rate by age
Primary source	**Vanguard "How America Saves"** — reports distribution activity by age
Backup source	**EBRI Retirement Security Projection Model (RSPM)** — models withdrawal behavior
Also	**Health and Retirement Study (HRS)** — longitudinal data on actual drawdown behavior for 50+ Americans

Variable	Detail
HRS URL	https://hrs.isr.umich.edu/data-products
Notes	Voluntary withdrawals before RMD age are lower than many expect. Many retirees live on Social Security + DB pension + taxable savings first, leaving tax-deferred accounts to grow. However, the transition from DB to DC means future cohorts will have less DB income and may withdraw earlier/more. For the model, we need age-specific withdrawal rates for ages 60–72 (pre-RMD) and excess withdrawal rates above RMDs for ages 73+.
Monte Carlo treatment	High uncertainty — this is one of the most uncertain parameters. Use a range from Vanguard data.

Panel 3.4 — TDF De-Risking Schedule (Equity Sold per Year per Dollar of AUM)

Variable	Detail
What	The annual change in equity allocation as a TDF ages along its glide path — this tells us how many dollars of equities are sold and replaced with bonds each year
Derivation	Computed from Panel 2.4 (the glide path). If equity allocation drops from 70% to 68% in one year, and the fund has $100B, then $2B of equities are sold.
Notes	This is not a separate data source — it's derived from the glide path. But it is a separate outflow channel that operates independently of RMDs. A TDF sells equities even if no money is withdrawn from the account.

Module 4: Net Flow Aggregation

No additional data needed

Module 4 is pure computation: sum inflows (Module 2) minus outflows (Module 3) across all age cohorts for each simulation year. The output is a distribution of annual net flows (in $billions) from 2025 to 2050.

Module 5: Price Impact

Panel 5.1 — Total U.S. Equity Market Capitalization

Variable	Detail
What	Annual total market cap of U.S. equities (for converting net flow dollars into % of market and applying the multiplier)

Variable	Detail
Primary source	**FRED — Wilshire 5000 Total Market Full Cap Index**
FRED series	WILL5000INDFC
URL	https://fred.stlouisfed.org/series/WILL5000INDFC
Backup	**World Federation of Exchanges —** monthly market cap data for NYSE + Nasdaq
Notes	Current total U.S. equity market cap is ~$55–60 trillion. For forward projection, we need an assumed growth rate. Use historical average nominal market return (~10%/year) as the baseline, with Monte Carlo draws around it. The multiplier applies to market cap: ΔMktCap = M × NetFlow.
Monte Carlo treatment	High uncertainty — market return is the most volatile parameter in the simulation. Use a distribution (e.g., lognormal) calibrated to historical mean and standard deviation.

Panel 5.2 — Passive Fund Share of U.S. Equities (Historical + Projected)

Variable	Detail
What	Time series of passive (index fund + ETF) share of total U.S. equity market, 2000–2024, plus Chinco & Sammon's true passive share estimate
Primary source	**ICI Factbook —** reports index fund and ETF AUM as % of total equity market annually
Also	**Chinco & Sammon (2024) —** Table/Figure data showing 33.5% true passive share as of 2021
Notes	ICI reports ~18% explicit index fund/ETF share as of 2023. Chinco & Sammon's 33.5% includes closet indexers and internal indexers. For projecting forward, we need an assumed growth rate. Historical trend: ~1–1.5 pp/year growth in reported passive share. We use this to project future passive share and compute the time-varying multiplier via Haddad's pass-through formula.
Monte Carlo treatment	Moderate uncertainty. Growth rate of passive share: 1–2 pp/year.

Panel 5.3 — Multiplier Estimates from the Literature

Variable	Detail
What	Not a dataset — these are parameter values from papers, already in hand
From Gabaix & Koijen	M ≈ 5 (central estimate). Micro: 0.3–2.5. Factor: 4.8–5.7. Macro: 1.5–6.5.

Variable	Detail
From Haddad et al.	$\chi = 3$ (strategic response). Pass-through ≈ 0.33. 11% more inelastic over 20 years.
Monte Carlo treatment	Baseline M drawn from a distribution centered on 5 with range 3–7 (our 2D scenario matrix).

Calibration & Context (Not Module-Specific)

Panel C.1 — S&P 500 Market Cap + Passive AUM Time Series (for descriptive chart)

Variable	Detail
What	Annual S&P 500 total market cap and total passive equity fund AUM, 2000–2024 — for the descriptive chart showing passive concentration growth
Source for S&P 500 market cap	**S&P Global** or **FRED WILL5000INDFC** (Wilshire covers broader market but S&P 500 is fine for illustration)
Source for passive AUM	**ICI Factbook** annual data
Notes	This is a descriptive/motivational chart for the introduction, not a model input. Shows the structural shift visually.

Panel C.2 — Historical Equity Returns (for backtesting and return distribution)

Variable	Detail
What	Annual total returns (with dividends) for the S&P 500 or Wilshire 5000, 1926–2024
Primary source	**FRED** (S&P 500 Total Return: SPXTR) or **Aswath Damodaran's data page** (NYU)
Damodaran URL	https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/histretSP.html
Notes	Needed to calibrate the return distribution for Monte Carlo draws. Historical mean ~10% nominal, ~7% real, $\sigma \approx 16$–18%.

Panel C.3 — Corporate Buyback Data (for discussion chapter)

Variable	Detail
What	Annual aggregate share buybacks for S&P 500 companies, 2000–2024

Variable	Detail
Primary source	**S&P Dow Jones Indices Buyback Quarterly Reports** (Howard Silverblatt)
Backup	**Federal Reserve Z.1 —** net equity issuance (negative = buybacks exceed issuance)
Notes	Recent annual buybacks ~$800B–$1T. This is for the discussion chapter (buybacks as moderator). We note the procyclical nature: buybacks collapsed in 2008–2009 and 2020.

Panel C.4 — International Demographic Synchronization (for discussion chapter)

Variable	Detail
What	Dependency ratios or MY ratios for major economies (EU, Japan, UK, China) to support the argument that international capital cannot fully offset U.S. demographic outflows because other countries are aging simultaneously
Primary source	**UN World Population Prospects 2024 —** provides age structure projections for all countries
URL	https://population.un.org/wpp/
Notes	Only needed for a few charts or a discussion paragraph. Not a model input. Japan and Europe are aging faster than the U.S.; China's demographic cliff is steeper. This counters the "foreign buyers will absorb the selling" argument.

Data Access Summary

Access Level	Sources
Free, direct download	Census Bureau, FRED, IRS Pub 590-B, BLS CPS, SSA AWI, UN WPP, Damodaran's page
Free but requires navigation	ICI Factbook (PDF, need to extract tables), Vanguard "How America Saves" (PDF), S&P Buyback Reports
May require institutional access	Morningstar TDF Landscape (check IE library), Fed SCF microdata (free but complex), HRS microdata (free but requires registration)
Already in hand (from papers)	Multiplier values (Gabaix-Koijen), χ and pass-through (Haddad), passive share (Chinco-Sammon), wealth decomposition (Poterba/Venti/Wise), participation rates (Madrian-Shea)

Priority Order for Tomorrow

Priority	Panel	Why First
1	1.1 (Census projections)	Foundation — everything depends on this
2	2.1 + 2.2 + 2.3 (Vanguard + BLS)	Inflow calibration — one PDF download gives participation + contribution + balances
3	3.1 (IRS RMD table)	Deterministic, takes 5 minutes
4	2.6 (ICI total DC assets/flows)	Calibration anchor — need this to validate modules 2 and 3
5	5.1 + C.2 (FRED market cap + returns)	Two quick FRED downloads
6	2.4 (TDF glide path)	Can use Parker et al. data initially, refine with Morningstar later
7	Everything else	Context, discussion, refinement